

Chapter 5

Name: _____

1. Define the following:

- a. Energy _____

- b. Potential Energy _____

- c. Kinetic Energy _____

- d. Solar Energy _____

- e. Thermal Energy _____

- f. Wind Energy _____

- g. Geothermal Energy _____

- h. Input Energy _____

- i. Converter _____

- j. Output Energy _____

- k. Chemical Energy _____

1. Law of Conservation of Energy _____

2. List 3 examples of kinetic energy. (pg 82)

3. List 3 examples of potential energy. (pg. 82)

4. List the energies you depend on when you are: (pg 82, 84, 85)

a. Walking to school

b. Driving to the store

c. Turning on the computer

d. Listening to the radio

5. List two ways the Sun's energy is converted for use. (pg 84, 85)

6. Is thermal energy a form of kinetic energy? Explain. (pg 84)

7. What energy conversions take place from the eating of an apple to the energy used to throw a ball? (pg 84, 85)

8. Briefly explain how plants and animals use energy from the Sun. (pg 84)

9. List 8 forms of energy and give 2 examples of each. (pg 85)

10. What 3 components are necessary for an energy conversion? (pg 86)

11. Identify the energy conversions that take place in the following scenarios.

a. When the inside of a car heats up when the car is left in the Sun.

b. When you paddle a canoe

c. A model rocket taking off

d. Drill press cutting a piece of wood

12. List 3 examples of energy conversion in which heat is given off. (pg 86 to 91)

13. Give two examples of waste thermal energy that might cause a problem? (Pg 91)

14. List 3 kinds of energy that come from a blow dryer. (pg 92)

15. Most of the energy produced by a car engine is not used to make the car move.
Where does the rest of the energy go? (pg 92)

16. Use an example to explain the Law of Conservation of Energy.(Pg 92)

17. Explain why input energy is greater than useful output energy. (pg 94)

18. Sketch two machines that have not been discussed but you are familiar with. On the sketch, identify where there is waste output energy. (pg 94)

Chapter 6 Generating Electric Energy

Name: _____

1. Define the following terms:

- a. Generator _____

- b. Turbine _____

- c. Hydroelectric _____

- d. Thermoelectric _____

- e. Thermonuclear _____

- f. Solar Cells _____

- g. Newton _____

- h. Power _____

- i. Watt _____

- j. Watt hour (W*H) _____

k. Kilowatt hour (kW*h) _____

l. Efficiency _____

2. Draw and label the main parts of a generator Pg 101

3. Describe how a simple generator produces electric energy. Pg 100/101

4. List two things you can do to increase the amount of electric energy produced by a generator. Pg 100/101

5. List 3 ways energy is generated in Canada. Pg 102

6. Describe how coal is used to produce electricity. pg 104
7. List two advantages of generating electric energy using a hydroelectric dam. Pg 105
8. What are some possible ecological damages that may result because of a hydroelectric dam? Pg 105
9. How much energy is lost as electricity moves from generating stations you your home? How is it lost? Pg 106

10. List 3 steps that take place as nuclear energy is converted to electric energy. Pg 106

11. What does a thermo-electric plant have in common with a thermonuclear power plant? Pg 106

12. What is power? pg 108

13. What are the units used to measure the power of electric appliances? Pg 109/110

14. Calculate the following examples using the formula $\text{Power} = \frac{\text{energy}}{\text{time}}$

REMEMBER time should be in seconds

a. 60 joules of energy are used in 2 hours

b. 120 joules in half a minute

c. 200 joules in 50 seconds

15. What are the units for the energy we buy from the power company. Pg 110

16. Using the formula $\text{energy} = \text{power} * \text{time}$ calculate the following examples.

a. A 100W light bulb is left on for 4 hours.

b. A 240 Watt fan is left on for 2 hours.

c. A 540 W stove is left on for 3 hours.

17. What is the difference between a kWh and a Wh? Pg 109/110

18. Why would the electric company use kilowatt hours instead of watt hours? Pg 110

19. How do we calculate efficiency? Pg 112

20. Why does an incandescent light bulb get so much hotter than a fluorescent light bulb? Pg 111

21. Use an example to define “efficiency”. Pg 111 to 113

22. State 3 ways to conserve energy at home. Pg 113

Chapter 7 Energy for Life

Name: _____

1. Define the following:

a. Chloroplasts _____

b. Chlorophyll _____

c. Photosynthesis _____

d. Cellular Respiration _____

e. Homeostasis _____

f. Metabolism _____

g. Carbohydrate _____

h. Glycogen _____

i. Fat _____

j. Saturated Fats _____

k. Unsaturated Fats _____

l. Protein _____

j. Metabolic needs _____

2. Where in plants does photosynthesis take place? Pg. 118

3. Write the formula for photosynthesis. Pg. 118

4. Where does cellular respiration take place? Pg 119

5. Write the formula for cellular respiration. Pg 119

6. What is the main source of energy of all the cells in our body? Pg 119

7. What would happen if we had no mitochondria in our bodies? Pg 119.

8. Why is cellular respiration important? Pg 119

9. What does lactic acid do to your muscles? Pg 119

10. How do animals capture energy from the sun? Pg 122

11. As you go higher on the pyramid of energy, is there more or less energy available? Why is this so? Pg 122

12. Why are there less organisms as you move up a food chain? Pg 122

13. How much energy is transferred to the next level in a food chain? Pg 122

14. Draw a food chain that is at least 4 animals long. Label the Producer, Primary consumer, Secondary consumer and the tertiary consumer. Pg 122

15. What is the internal balance in the body called? Pg 123

16. List 2 things that your body will do to cool itself down? Pg 123

17. List 2 things your body does to warm itself up? Pg 123

18. List 2 adaptations that the Inuit have to keep them warm. Notes

19. List 2 adaptations that desert people have that allows them to survive. Notes

20. How does homeostasis and metabolism work together to keep an organism alive? Pg 124

21. What is the main source of energy for the human body? Pg 125

22. What role do carbohydrates play in cellular respiration? Pg 125

23. What percent of our daily calories should be carbohydrates? Pg 125

24. What is the difference between glycogen and glucose? How do they work together? Pg 125

25. List 4 carbohydrates Pg 125

26. What do we need fat for? Pg 126

27. List 4 sources of fat. Pg 126

28. Complete the following flow chart using the following words.

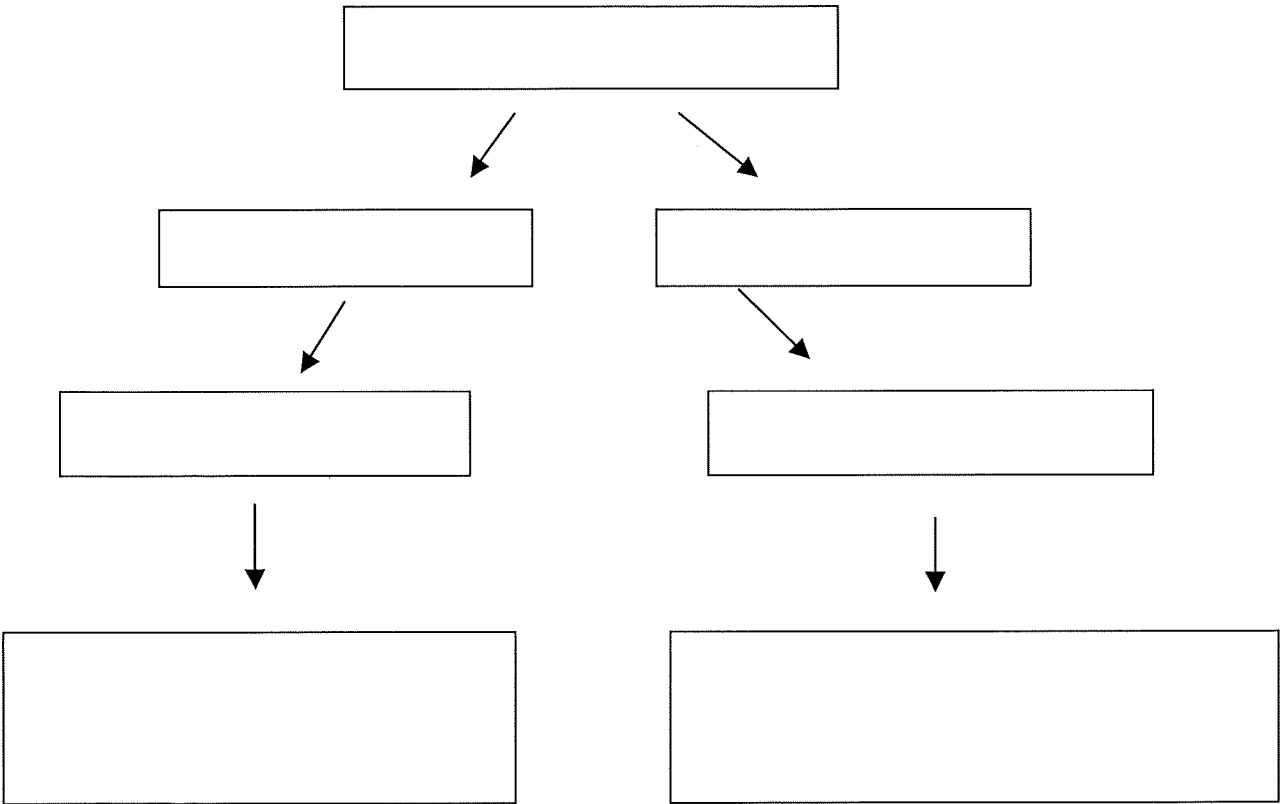
Fats
Unsaturated

Decreased risk of heart disease

Saturated

Animal Fats

Increased risk of heart disease



29. What is the difference between protein and carbohydrates? Pg 127

30. Why is protein important for our body? Pg 127

31. List 4 sources of protein. Pg 127

32. Why do children need more protein than adults? pg 127

33. In what situations would your body use protein for energy? Pg 127

34. In your own words define metabolism? 130/131

35. Describe some of the differences in nutrient and energy requirements between a baby and a senior citizen? Pg 130

36. List 3 occupations in which you would expect workers to have high metabolisms? Pg 131

37. For each of the following situations, predict whether the person's metabolic rate will go up, down, or stay the same.

a) A construction worker resigns and takes on a rural mail route.

b) A mother stops breast feeding.

c) A high school gymnast begins training for the provincial finals

d) A weight lifter takes time off from training

Chapter 8

Name: _____

1. Define the following:

- a. Fossil fuels _____

- b. Coal _____

- c. Natural gas _____

- d. Crude oil _____

- e. Petroleum _____

- f. Reservoir rock _____

- g. Seismic survey _____

- h. Pump jack _____

- i. Lift pump _____

- j. Distillation _____

- k. Combustion _____

l. Hydrocarbons _____

m. Non-renewable _____

n. Acid deposition _____

o. Global climate change _____

2. List 3 fossil fuels.(pg 137)

3. Sketch and LABEL four steps in the formation of coal. (pg 136)

4. Describe two ways in which coal is mined. (pg 138)
5. List 3 uses for each of the fossil fuels.(pg 137)
6. Explain how the various components of crude oil can be separated. (Pg 143)
7. Sketch how oil, water and natural gas arrange themselves under the Earth's surface. (pg 140)

8. Explain how shockwaves are used to find oil.(pg 140)
9. How are pumps used to bring oil to the surface?(pg 141)
10. List 3 technologies that use combustion to convert the energy stored in fossil fuels into other forms of energy. (pg 138 to 143)
11. Copy the word and chemical equations for the combustion of propane. $C_3H_8(g)$. Label the input energy, converter, and output energy. (Pg 144)

12. Identify the two gases produced by both combustion and cellular respiration. (pg 146)

a. How else is cellular respiration similar to the combustion of a fossil fuel? (pg 146)

13. Why are fossil fuels said to be non-renewable? (pg 147)

14. List 3 risks associated with our use of fossil fuels.(pg 147/149)

15. Use a labeled diagram or flowchart to explain how an increase in greenhouse gas might make Earth warmer. (pg 149)

16. Why is global climate change a concern?(pg 149)

